

MANOKOTAK, AK, USA

WMO: 703055

Lat: 58.990N

Lon: 159.050W

Elev: 16

StdP: 101.13

Time Zone: -9.00 (NAK)

Period: 06-19

WBAN: 00109

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.1	(c) -23.8	(d) -30.3	(e) 0.2	(f) -26.1	(g) -27.6	(h) 0.3	(i) -23.1	(j) 15.8	(k) -2.7	(l) 14.7	(m) -5.5	(n) 0.9	(o) 30	(p) 0.771

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.9	(c) 22.1	(d) 15.5	(e) 20.0	(f) 14.3	(g) 18.2	(h) 13.4	(i) 16.2	(j) 20.3	(k) 15.1	(l) 18.7	(m) 14.1	(n) 17.2	(o) 3.9	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.3	10.2	17.3	13.2	9.5	16.0	12.5	9.1	15.2	45.0	20.2	41.9	18.6	39.5	17.2	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.3	11.0	9.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-28.7	18.1	3.9	1.8	-31.5	19.4	-33.8	20.4	-36.0	21.4	-38.9	22.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-8.3	-6.6	-6.2	-0.2	6.6	10.9	12.7	12.5	9.1	3.8	-4.2	-7.2
	DBStd	9.51	9.04	6.48	5.87	4.81	3.40	2.58	2.81	2.20	2.90	3.68	6.01	7.75
	HDD10.0	3183	567	466	502	305	116	18	5	4	48	193	427	533
	HDD18.3	5983	826	699	760	555	365	224	176	182	277	451	677	791
	CDD10.0	244	0	0	0	0	9	44	89	81	21	1	0	0
	CDD18.3	2	0	0	0	0	0	0	2	0	0	0	0	0
	CDH23.3	22	0	0	0	0	2	5	16	0	0	0	0	0
	CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0
(14) Wind	WSAvg	3.9	4.8	4.8	3.8	4.1	3.8	3.4	3.4	3.3	3.4	4.2	3.9	4.6
(15) Precipitation	PrecAvg	843	70	58	53	42	49	57	74	82	101	86	87	82
	PrecMax	1220	164	116	154	78	106	93	123	133	202	158	196	184
	PrecMin	540	16	10	4	7	7	28	37	34	21	8	12	17
	PrecStd	154	32	28	36	19	23	15	21	27	39	33	42	35
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	3.6	6.3	12.2	22.6	22.9	25.5	22.2	18.9	12.4	7.5	4.3
		MCWB	3.7	2.3	3.9	6.0	12.3	14.7	18.7	15.7	13.2	9.4	6.1	3.0
	2%	DB	3.4	2.5	4.1	10.1	18.0	20.2	22.6	20.3	17.2	11.0	6.1	3.1
		MCWB	2.3	1.5	2.6	5.6	10.9	13.5	16.4	14.9	12.3	9.1	5.2	2.1
	5%	DB	2.6	1.8	2.5	7.7	15.0	18.4	20.7	18.6	15.1	9.9	4.3	2.4
		MCWB	1.7	1.1	1.3	4.3	9.3	12.9	15.5	13.9	11.9	8.2	3.6	1.6
	10%	DB	1.9	1.2	1.4	6.0	12.6	16.5	18.3	17.1	13.5	8.8	2.9	1.5
		MCWB	1.1	0.6	0.5	3.5	8.0	11.7	14.2	13.4	11.0	7.6	2.4	1.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	2.8	4.5	6.8	13.2	15.9	19.6	16.7	14.8	10.5	6.9	3.5
		MCDB	5.4	3.3	5.8	10.3	20.0	20.7	26.2	20.4	16.9	11.2	7.3	4.3
	2%	WB	2.6	1.7	2.8	5.7	11.4	14.3	17.0	15.4	13.3	9.5	5.3	2.5
		MCDB	3.2	2.2	3.9	9.3	17.6	19.0	21.3	18.8	15.6	10.5	5.6	3.0
	5%	WB	1.8	1.2	1.5	4.7	9.7	13.2	15.6	14.6	12.1	8.5	3.9	1.7
		MCDB	2.4	1.6	2.2	7.5	14.2	17.2	19.8	17.4	14.1	9.3	4.5	2.2
	10%	WB	1.1	0.6	0.6	3.7	8.4	12.2	14.6	13.8	11.3	7.8	2.6	1.1
		MCDB	1.7	1.0	1.3	5.6	11.8	15.8	18.1	16.2	13.2	8.5	3.1	1.5
(35) Mean Daily Temperature Range	MDBR		6.0	6.9	7.6	6.8	7.9	8.2	6.9	6.6	6.8	5.8	6.2	5.8
	5% DB	MCDBR	3.7	4.8	5.6	8.6	12.6	11.7	11.4	10.5	9.5	5.4	5.2	4.6
		MCWBR	3.1	4.2	4.5	5.2	7.0	6.3	6.4	5.8	5.8	4.0	4.6	4.2
	5% WB	MCDBR	3.6	4.7	5.1	8.0	11.5	10.6	10.6	8.4	8.1	4.6	4.8	4.4
		MCWBR	3.1	4.3	4.2	4.9	6.6	6.0	6.2	5.0	5.4	4.0	4.5	4.2
(40) Clear-Sky Solar Irradiance	taub		0.258	0.303	0.319	0.367	0.402	0.367	0.365	0.369	0.332	0.308	0.289	0.262
	taud		2.125	2.156	2.145	2.127	2.193	2.404	2.451	2.433	2.513	2.416	2.104	1.986
	Ebn at Noon		547	699	809	829	825	867	861	828	810	710	479	369
	Edh at Noon		47	78	105	128	130	107	100	95	75	60	48	38
(44) All-Sky Solar Radiation	RadAvg		0.41	1.03	2.80	4.38	5.18	5.29	4.50	3.63	2.54	1.32	0.53	0.23
	RadStd		0.08	0.20	0.45	0.53	0.71	0.45	0.48	0.44	0.33	0.16	0.08	0.03

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page